Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 8 [2023 P1]

Test Date: 30th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

MATHEMATICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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1. There are 20 points $A_1, A_2, A_3, \ldots\ldots\ldots\ldots, A_{20}$ lying on a circle, distance between two consecutive points is same. Three points are selected at random. X represent no. of ways to select them. Y represent no. of ways to select them such that no two adjacent objects are selected and Z represent no. of ways to select such that chosen objects are not diametrically opposite then

A) X = 1140 **B)** Y = 800 **C)** Z = 960 **D)** $Y \cap Z = 660$

2. An ellipse having eccentricity $\frac{3}{4}$ passes through the point P(5, 4). If the nearest focus is (1, 1) to and equation of tangent at P on the ellipse 4x + 3y – 32 = 0, then:

A) Length of latus rectum of ellipse is 17.5

B) Length of chord through (1, 1) parallel to the tangent at P is 17.5

C) Length of perpendicular from the focus other than (1, 1) on the tangent at P is 35

D) The point of intersection of perpendicular tangents drawn to ellipse always lies on the circle $(x+1)^2 + (y+8)^3 = 575$

3. If $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, then which of the following is/are INCORRECT (where A^{-1} is inverse of matrix A, I is an unit matrix of order 3, adj A is adjoint of matrix A, |A| denotes determinant value of matrix A).

A) $\frac{1}{81}\left|\text{adj}(2A).\text{adj}\left(3A^{-1}\right).A^{-1}\right|$ is equal to 288

B) $\left(A^{-1}\right)^2 = \frac{1}{2}\left(A - 5I + 7A^{-1}\right)$

C) $\left|A^5 - 2A^4\right|$ is equal to 16 **D)** $A^{-1} = \frac{1}{2}\left(A^2 - 5A + 6I\right)$

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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4. Consider $f:[0,\infty)\to[0,\infty)$ is bijective function and it satisfies $f(x).e^{f(x)}=x$, then the value of $\lim\limits_{x\to\infty}\dfrac{f(x)}{\ln x}$ is :

A) 0 **B)** 1 **C)** e **D)** does not exist

5. Real number x, y, z satisfies

$x + xy + xyz = 1$

$y + yz + xyz = 2$

$z + xz + xyz = 4$

If the largest possible value of xyz is $\dfrac{a+\sqrt{b}}{c}$, where b and c are prime numbers, then (a + b + c) equals :

A) 20 **B)** 22 **C)** 24 **D)** 26

6. Let for a sequence $a_n=\dfrac{1}{\sqrt{n}}$, $n\in N$ then $\left[\sum\limits_{n=1}^{80}a_n\right]$ is equal to (where [.] represent greatest integer function) :

A) 16 **B)** 17 **C)** 18 **D)** 15

7. Let $F_n(x)=\int\sin(n+2)x\,\sin^n x\,dx$ and $F_n(\pi)=0$ where $n\in N$. If the value of $F_{16}\left(\dfrac{3\pi}{2}\right)$ is a, then determine $\left(\dfrac{1}{a}+2\right)$

A) 19 **B)** 25 **C)** 30 **D)** 28

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. If $\sum\limits_{r=0}^{3n}a_r(x-4)^r=\sum\limits_{r=0}^{3n}A_r(x-5)^r$ and $a_k=1\ \forall k\ge 2n$ and $\sum\limits_{r=0}^{3n}d_r(x-8)^r=\sum\limits_{r=0}^{3n}B_r(x-9)^r$ and $\sum\limits_{r=0}^{3n}d_r(x-12)^r=\sum\limits_{r=0}^{3n}D_r(x-13)^r$ and $d_k=1\ \forall k\ge 2n$, then the value of $\dfrac{A_{2n}+D_{2n}}{B_{2n}}$

9. If θ is the angle between $\bar{a}$ and $\bar{b}$ such that $\sqrt{3}(\vec{a}.\vec{b})+\left|\vec{b}-(\vec{a}.\vec{b})\vec{a}\right|=1+\vec{b}.\vec{b}$ where $|\bar{a}|=1$ then $\operatorname{cosec}^2\theta$ is

10. If $f(x)=\left(\left[\{x\}\right]\tan^{-1}\left(\frac{x^2-3x-1}{x^2-3x+5}\right)+3-x^7\right)^{\frac{1}{7}}$

Where [k] and {k} denotes greatest integer and fractional part of 'k' respectively, then the value of $f^{-1}(50)-f(50)+f(f(100))$, is

11. Suppose α, β, γ are complex numbers satisfying the following system of the equations $\alpha+\beta+\gamma=6, \alpha^3+\beta^3+\gamma^3=87$ and $(\alpha+1)(\beta+1)(\gamma+1)=33$.

If $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{m}{n}$ (where m and $n \in N$ are relatively prime), then the value of $\left[\frac{n}{m}\right]$ is (where [.] denotes greatest integer function)

12. If $2|x^2-4|=x^2-4x+a$ has exactly three distinct solutions then sum of all possible values of 3a is ……

13. Let $f(x)=1-|x-2|$ for $1\le x\le 3$ and $f(3x)=3f(x)$ for all positive real 'x'. The number of points in (1,100) where f(x) is non differentiable, is

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14. Match the following Column – I with Column – II:

	COLUMN-I		**COLUMN-II**
P)	Let $f(x)=x^2\sin\frac{\pi}{x}(x\neq 0)$, then the number of points in $\left(\frac{1}{21},1\right)$ at which $f'(x)=0$ is	**1)**	18
Q)	Let $f(x)=x^2+px+q+3$, if $f(f(1))=f(f(-2))=0$ where $f(1)\neq f(-2)$ and $g(x)=x^2+(8p^2-18p)x-q+2$, them $2g(2)$ is	**2)**	20
R)	Let $f(x)=x^9-3x^8-54x^7+x^4-8x^3-8x^2-8x+9$ Then f(9) will be equal to	**3)**	23
S)	Let $L=\lim_{x\to 0}\left(1+e^{-\frac{1}{x^2}}.\tan^{-1}\frac{1}{x^2}+x^2.e^{-\frac{1}{x^2}}.\sin\frac{1}{x^4}\right)^{e^{\left(\frac{1}{x^2}\right)}}$, then	**4)**	27

	$\left[15\log_e L\right]$ is (where [.] represents greatest function)		

The correct option is:

A) P – 2; Q – 4; R – 1 ; S – 3 **B)** P – 4; Q – 3; R – 2; S – 1

C) P – 4; Q – 3; R – 1 ; S – 2 **D)** P – 2; Q – 4; R – 2 ; S – 3

15. Match the statements of Column – I with values of Column – II:

	COLUMN-I		**COLUMN-II**
I)	Q is image of P(2, 4) in bisector line L of 1st quadrant, S is circle with centre Q and radius 4. A circle M is drawn such that it touch line L is (1, 1) and its radical axis with S passes through P then its radius is	**P)**	$12\sqrt{2}$
II)	Tangents are drawn from $P(K,0)$ on $x^2 + y^2 = 50$, meeting y-axis at Q and R possible value of 'K' for which area of ΔPQR is minimum	**Q)**	$\sqrt{50}$
III)	If length of major axis of ellipse $2x^2 + xy + y^2 = 5$ is 2k and $k^2 = \frac{10}{a-b}$, then a^2b is : (a is integer)	**R)**	$\frac{\sqrt{162}}{2}$
IV)	Centre of a circle passing through foci of $\frac{x^2}{25} - \frac{y^2}{24} = 1$ and Through the points of intersection of tangents at vectors with Tangent at point $\left(5\sqrt{2}, 2\sqrt{6}\right)$ is (h, k) then $\left\|2.\sqrt{3}k\right\|$ equal	**S)**	10
		T)	$\sqrt{162}$

A) I – Q; II – R; III – P ; IV – T **B)** I – R; II – S; III – T; IV – P

C) I – R; II – P; III – T ; IV – Q **D)** I – Q; II – S; III – S ; IV – P

16. A box contains 30 tickets, each with a different number from 1 to 30. Four tickets are drawn at random without replacement. Let p, q, r, s be the number on the ticket drawn.

	COLUMN-I		**COLUMN-II**
I)	The probability that min (p, q, r, s) < 15 is	**P)**	$\frac{19}{609}$
II)	The probability that the second smallest number is 20	**Q)**	$\frac{248}{261}$
III)	The probability that the sum of p + q + r + s is even	**R)**	$\frac{103}{609}$
IV)	The probability that product 'pqrs' is even	**S)**	$\frac{731}{783}$
		T)	$\frac{2951}{5481}$

A) I – P; II – S; III – Q ; IV – R **B)** I – P; II – S; III – R; IV – Q

C) I – S; II – P; III – T ; IV – R **D)** I – S; II – P; III – T ; IV – Q

17. Match the following Column – I with Column – II

	COLUMN-I		**COLUMN-II**
I)	The value of $\int_0^1 \left(\sqrt[2021]{1-x^{2022}} - \sqrt[2022]{1-x^{2021}} \right) dx$	**P)**	4
II)	If $I = \frac{\int_0^{51} [x+25]dx}{\int_0^{51} \{x+25\}\,dx}$, then $\frac{1}{25}$ is equal to (where [.] → G.I.F and {.} → F.P.F)	**Q)**	0
III)	Let $L = \lim_{n\to\infty} \frac{1}{n}\left(\sin^2\frac{\pi}{2n} + \sin^2\frac{2\pi}{2n} + ... + \sin^2\frac{n\pi}{2n} \right)$, then 2L is equal to	**R)**	2
IV)	If $J = \int_0^{\infty} xe^{-2x}\,dx$ then 8J is equal to	**S)**	3
		T)	1

A) I – P; II – Q; III – R ; IV – S **B)** I – Q; II – P; III – R; IV – T

C) I – Q; II – P; III – T ; IV – R **D)** I – P; II – Q; III – S ; IV – R

PHYSICS **Max Marks: 60**

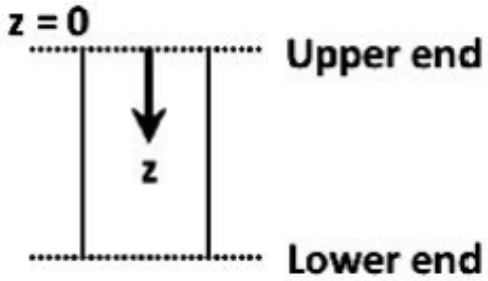
SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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18. In an organ pipe (which may be closed or open) of 99 cm length, standing wave is set up whose equation is given by longitudinal displacement. x = 0.3 (mm) cos $\left[(2\pi / 0.8)(z + 0.01\text{m})\right]$ cos 400t, where z is measured form the top of tube in metres and 't' is in seconds. Take end correction = 1 cm. Choose the CORRECT statements.



A) The upper end and lower end of tube are open and closed respectively.

B) The upper end and lower end of tube are closed and open respectively.

C) The air column is vibrating in third harmonic.

D) The air column is vibrating in second overtone.

19. In an AC circuit, we have an inductance L joined in series to two resistors of resistance R each. In the first case, the resistors are in series with each other and in the second case, the resistors are parallel to each other. The power consumed by the circuit in the first case is 810W and in the second case, it is 506.25 W. If the source voltage is 225 V,

A) Resistance R = 20 Ω

B) In the first case, the phase difference between current through the source and voltage across it is 37^0.

C) In the second case, the phase difference between current through the source and voltage across it is $\tan^{-1}(3)$

D) Inductive reactance is 30Ω.

20. Light of wavelength λ is incident on a spherical zinc ball of radius 10m. The ball has a charge of 1 nC. The ball is in vacuum. The work function of zinc is 4 eV.

A) If λ = 250 nm, electrons will come out of the ball but all of them will re-enter the ball.

B) If λ = 350 nm, electrons will not come out of the ball.

C) If λ = 300 nm, electrons will come out of the ball but all of them will re-enter the ball.

D) If λ = 155 nm, electrons will come out of the ball but some of them will never re-enter the ball

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

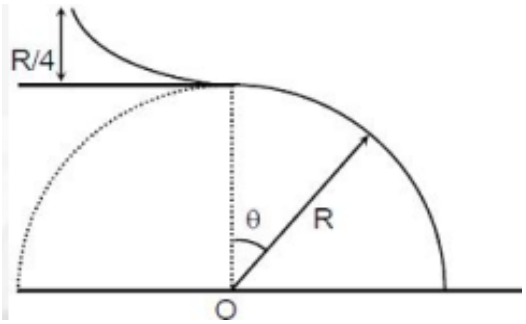
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21. A skier plans to ski a smooth fixed hemisphere of radius R. He starts from rest from a curved smooth surface of height (R/4). The angle θ at which he leaves the hemisphere is:



A) $\cos^{-1}\left(\frac{2}{3}\right)$ **B)** $\cos^{-1}\left(\frac{5}{\sqrt{3}}\right)$ **C)** $\cos^{-1}\left(\frac{5}{6}\right)$ **D)** $\cos^{-1}\left(\frac{5}{2\sqrt{3}}\right)$

22. One mole of an ideal diatomic gas undergoes a thermodynamic process, in which its molar heat capacity varies directly proportional to temperature as $C = \alpha T$, where α is a positive constant. Work done by the gas when it is heated from initial temperature T_0 to a final temperature$3\,T_0$ will be

A) $4\alpha T_0^2$ **B)** $\left(\alpha T_0 - R\right)\frac{3T_0}{2}$ **C)** $\left(4\alpha T_0 - 5R\right)T_0$ **D)** $\left(3\alpha T_0 - 5R\right)\frac{T_0}{2}$

23. A point source S of light is placed at a depth 'd' below the surface of water in a large and deep lake. Maximum fraction of light that escapes inspace above directly from water (refractive index = μ) surface is given by

A) $\frac{1}{2}-\sqrt{\frac{\mu^2-1}{\mu}}$ **B)** $\sqrt{\frac{\mu^2-1}{\mu}}$ **C)** $\frac{1}{2}-\frac{\sqrt{\mu^2-1}}{2\mu}$ **D)** $\frac{1}{2}-\frac{\sqrt{\mu^2-1}}{\mu^2}$

24. A total charge Q is uniformly distributed over a non-conducting disc of radius r. There is a time varying magnetic field perpendicular to its plane and changing at the uniform rate of $\frac{dB}{dt}$. The magnitude of torque experienced by the disc is

A) $\frac{Qr^2}{2}\left(\frac{dB}{dt}\right)$ **B)** $\frac{Qr^3}{3}\left(\frac{dB}{dt}\right)$ **C)** $\frac{1}{4}Qr^2\left(\frac{dB}{dt}\right)$ **D)** $\frac{1}{2}Qr^3\left(\frac{dB}{dt}\right)$

SECTION-III

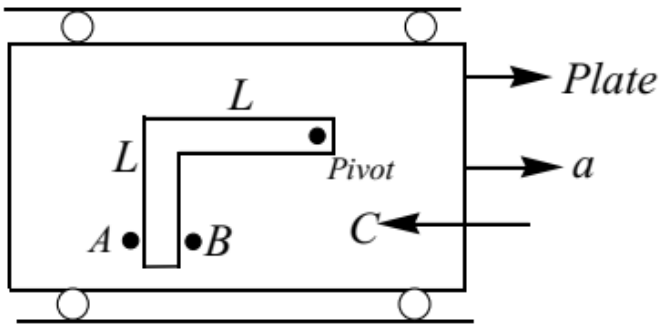
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

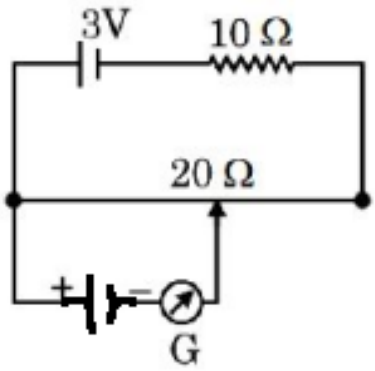
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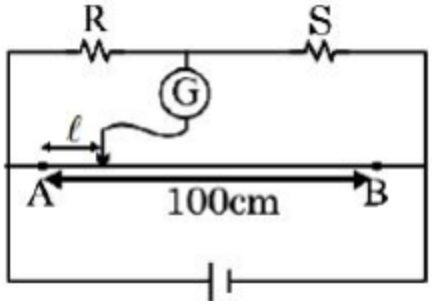
25. An L shaped thin uniform rod of mass M and total length 2L is freely hinged to vertical plane at C. The bar is prevented from rotating by two pegs A and B fixed to the plate. If the acceleration of plate is given by 'a' (*in m/*s^2) for which no force is exerted on the either peg A or B. Find the value of $\frac{a}{10}$? $\left(g=10\ m/s^2\right)$



26. A 10 m long potentiometer wire has a resistance of 20 ohm. It is connected in series with a battery of emf 3V and a resistance of $10\,\Omega$. The internal resistance of cell is negligible. If the length can be read accurately up to 1 mm, the potentiometer can read voltage with an accuracy of $\frac{k}{10}$ (in milli volt). Then the value of 'k' is



27. Consider the meter bridge circuit without neglecting end corrections. If we used $100\,\Omega$ and $200\,\Omega$ resistance in place of R and S respectively, we get null deflection at $l_1 = 33.0$ cm. If we interchange the resistances, the null deflection was found to be at $l_2 = 67.0$ cm. If α and β are the end correction, then the value of $\alpha + \beta$ should be

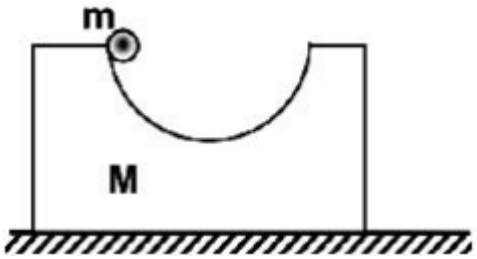


28. An alternating sinusoidal current of frequency $\omega = 1000\ \text{s}^{-1}$ flows in the winding of a straight solenoid whose cross-sectional radius is equal to R = 6.0 cm. Find the ratio of peak values of electric and magnetic energies within the solenoid. If your answer is $\text{n} \times 10^{-15}$, find n = ?

29. Two points, A and B, are located on the ground a certain distance $\text{d} = 10\sqrt{2}$ m apart. Two rocks are launched simultaneously from points A and B with equal speeds but at different angles. Each rock lands at the launch point of the other. Knowing that one of the rocks is launched at an angle $\theta = 37^0$ with the horizontal, what is the minimum distance between the rocks during the flight? (given cos $37^0 = 4/5$)

30. A semi-cylinder is cut out of a block and the block is kept on the horizontal surface. A small body of mass 'm' is released at the top of the cylinder shaped hole. (Friction is negligible.)

If the block is able to move on the horizontal surface without friction, then the force exerted on the block by the small body at the lowest point is 7mg/2. Find the ratio of M/m?



SECTION – IV
(MATCHING TYPE)

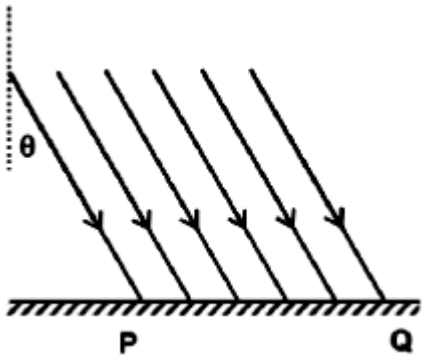
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
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Negative Marks: –1 In all other cases.

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31. A parallel beam of light is incident on a fixed surface PQ at an angle θ with the vertical as shown in the figure. The intensity of light is I and area of surface PQ is A. In Column-I nature of reflection are given and in Column-II force and generated radiation pressure are given. Then match Column-I with suitable option Column -II. (Given that $I = 500\ W/m^2,\ A = 1.5\,m^2,\ \theta = 37^0$)

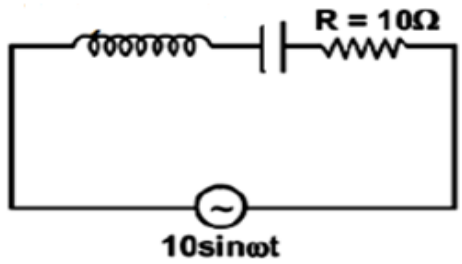
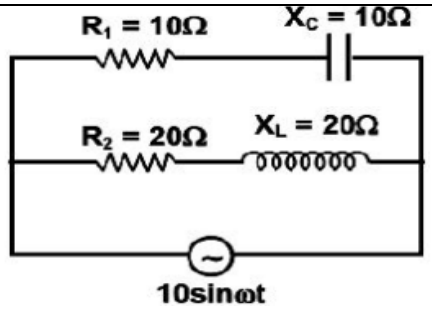
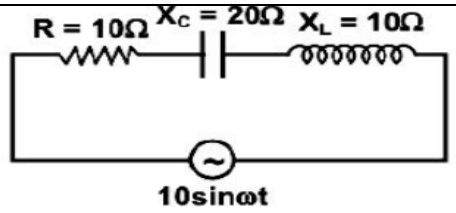
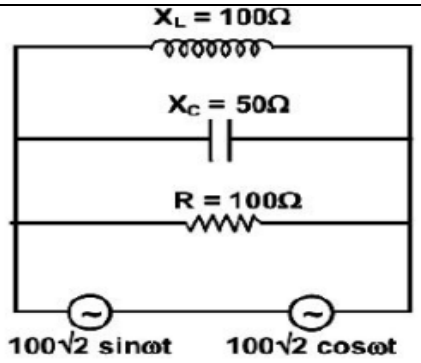


	COLUMN-I		COLUMN-II
P)	If all the incident energy is absorbed by surface PQ then force on the surface and radiation pressure generated are respectively.	**1)**	3.2×10^{-6} N, 2.13×10^{-6} N / m^2
Q)	If all the incident energy is reflected by surface PQ then force on the surface and radiation pressure generated are respectively.	**2)**	2.24×10^{-6} N, 1.38×10^{-6} N / m^2
R)	If 30% incident energy is reflected by the surface then force on the surface and pressure generated are respectively. (Rest absorbed)	**3)**	2.74×10^{-6} N, 1.81×10^{-6} N / m^2
S)	If 70% incident energy is reflected back then force on the surface and radiation pressure are respectively. (Rest absorbed)	**4)**	2×10^{-6} N, 1.06×10^{-6} N / m^2
		5)	4.2×10^{-6} N, 3.13×10^{-6} N / m^2

A) P – 4; Q – 1; R – 3; S – 2 **B)** P – 3; Q – 2; R – 1; S – 4

C) P – 3; Q – 5; R – 4; S – 2 **D)** P – 4; Q – 1; R – 2; S – 3

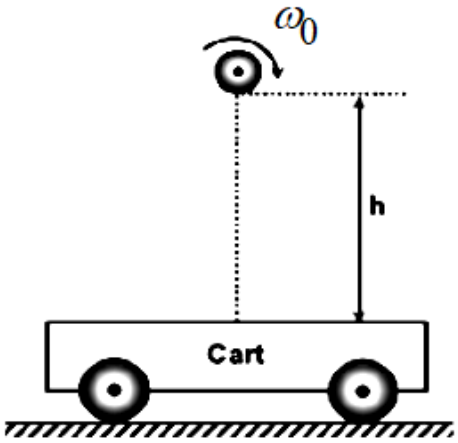
32. Some electric circuits with an A.C. source is given in Column-I. In Column -II instantaneous current and average power delivered by A.C. source are given. The Match Column -I with Column -II

	COLUMN-I		**COLUMN-II**
P)	$X_L = \frac{1}{2}\Omega X_C = \frac{1}{2}\Omega$ 	**1)**	$I = \frac{\sqrt{10}}{4}\sin\left[\omega t + \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2}\right)\right]$. $P = \frac{15}{4}$ watt
Q)		**2)**	$I = 2\sqrt{2}\cos\omega t$, $P = 200$ watt
R)		**3)**	$I = \sin\omega t$, $P = 5$ watt
S)		**4)**	$I = \frac{1}{\sqrt{2}}\sin\left(\omega t + \frac{\pi}{4}\right)$, $P = \frac{5}{2}$watt
		5)	$I = \frac{\sqrt{10}}{4}\sin\left(\omega t + \frac{\pi}{4} - \tan^{-1}(2)\right)$, $P = \frac{7}{2}$watt

A) P – 4; Q – 4; R – 1; S – 4 **B)** P – 3; Q – 3; R – 1; S – 3

C) P – 3; Q – 1; R – 4; S – 2 **D)** P – 4; Q – 3; R – 4; S – 1

33. A solid sphere of mass m = 80 kg and radius r = 0.2 m is released from height h = 5/4 meter. Sphere is initially rotating about horizontal axis passing through its centre of mass. It hits with a stationary cart of mass M = 200 kg exactly at the centre of cart. The cart can move smoothly on the horizontal surface. The collision between sphere and cart occurs in such a way that sphere reaches at same vertical displacement after collision and falls back onto it again. It is found that sphere starts pure rolling at the end of first collision. The coefficient of friction between sphere and cart is $\mu = 0.1$. Match the statement given in Column-I to the values given in Column-II. $\left(g = 10\,m/sec^2\right)$

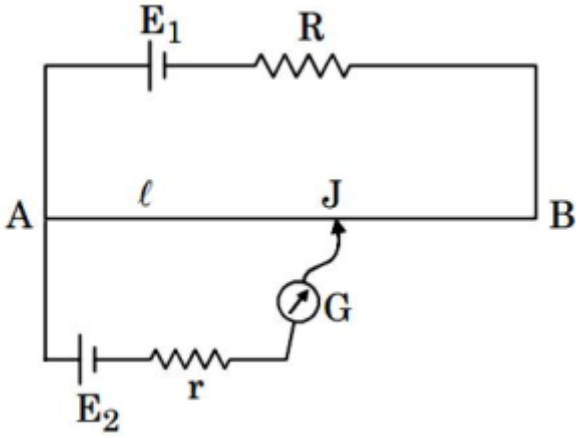


	COLUMN-I		COLUMN-II
P)	The minimum length (in meter) of cart to occur second collision with the sphere	**1)**	172
Q)	Initial angular velocity ω_0 (in rad/sec) of sphere on the cart during the process.	**2)**	2.8
R)	Magnitude of work done (in Joule) by sphere on the cart during the process	**3)**	156
S)	Magnitude of work done (in Joule) by cart on the sphere during the process	**4)**	19.4
		5)	16

A) P – 2; Q – 4; R – 5; S – 1 **B)** P – 1; Q – 3; R – 1; S – 3

C) P – 3; Q – 5; R – 4; S – 2 **D)** P – 4; Q – 3; R – 4; S – 1

34. In the potentiometer arrangement shown in figure, null point is obtained at length '*l*'. Match the following (E_2 is ideal)



	COLUMN-I		COLUMN-II
P)	If E_1 is increased	**1)**	*l* should increase
Q)	If R is increased	**2)**	*l* should decrease
R)	If E_2 is increased	**3)**	*l* should remain the same to again get null point
S)	If 'r' is shunted	**4)**	*l* becomes zero

A) P – 1; Q – 2; R – 3; S – 4 **B)** P – 2; Q – 1; R – 3; S – 4

C) P – 2; Q – 1; R – 1; S – 3 **D)** P – 4; Q – 1; R – 2; S – 3

CHEMISTRY **Max Marks: 60**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

35. Consider three-dimensional hexagonal close packing of identical rigid spheres. Two different types of voids are formed. Which of the following statements is/are correct about these voids?

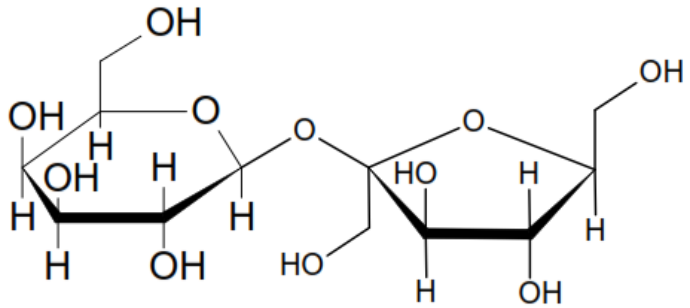
A) There are two different types of tetrahedral voids in terms of volume of the void.

B) All the depressions of first layer are not completely covered by the spheres of the second layer to form tetrahedral void at each depression.

C) There is only one type of octahedral voids in terms of volume of the void.

D) Centres of tetrahedral voids and octahedral voids are at same distance from the plane of centres of spheres of one of the layers arranged in hexagonal close packing.

36. Correct statement/s regarding the following sugar is/are



A) It forms galactose and fructose on hydrolysis.
B) It is a reducing sugar

C) It can exist in another anomeric form.

D) It contains β-glycosidic linkage

37. Which of the following order is/are CORRECT for indicated property?

A) Pd < Pt (Maximum number of electrons having $l = 2$)

B) 1st period < 2nd period < 3rd period (Number of s-block elements)

C) C < N < O (Z_{eff} on last electron)

D) Cr < Mn (unpaired electrons)

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

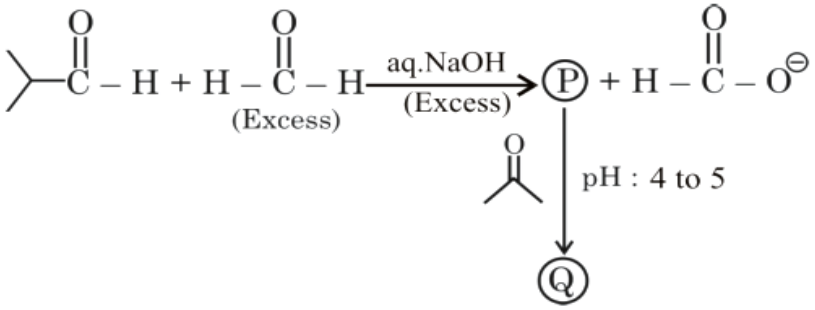
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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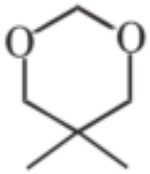
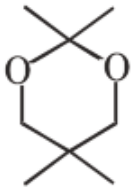
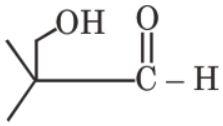
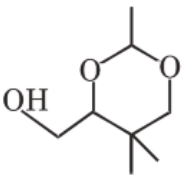
38. The p^{kb} value of NH_4OH is 4.75. An aqueous solution of NH_4OH is titrated with HCl. The pH of the solution at the point where Half of NH_4OH has been neutralized will be

A) 9.25 **B)** 8.25 **C)** 7.50 **D)** 4.75

39.



Q is

A)  **B)**  **C)**  **D)** 

40. Metal salts [X] when heated with borax, the following observation are made. Colour of the bead in oxidizing flame in hot and cold is violet, where as colour of the bead in reducing flame is colourless in hot as well as in cold conditions. The metal salt is a salt of

A) Mn^{2+} **B)** Cu^{2+} **C)** Cr^{3+} **D)** Ni^{2+}

41. In the extraction of the aluminium by Hall-Heroult process, identify the incorrect statement

A) The oxygen gas liberated at the anode reacts with graphite anode at 950^0 C gives $CO \,\&\, CO_2$

B) A bunch of Graphite rods are act as anode and steel vessel inside lined with silica act as Cathode.

C) Cryolite or CaF_2 lowers fusion temperature and increases conductivity

D) Aluminium is heavier than cryolite

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

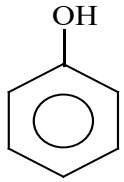
Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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42. The oxidation state of Cr in K_3CrO_8 is

43. Rate of diffusion of ozonized oxygen is $0.4\sqrt{5}$ times that of pure O_2. When the percentage of association is $x \times 10$ what is 'x'. assuming ozonized oxygen is formed from pure O_2

44. Butyric acid $\xrightarrow[(ii)\ Cr_2O_3, 780K,\ 20\ atm]{(i)\ NaOH,\ electrolysis} P(product)$. Number of sp^2 carbon atoms present in product "P" is:

45. 

$$\xrightarrow[(iv)\ AlCl_3\ (v)\ NaBH_4 / EtOH]{(i)\ NaOH\ (ii)\ CO_2\ (iii)\ H^+ / Ac_2O} P\ (major)$$

Total no of oxygen atoms in the aromatic product (P) is

46. Consider following complex formation reaction.

I) $Co^{3+}_{(aq)} + KCN_{(excess)} \rightleftharpoons$ complex ion P

II) $Ni^{2+}_{(aq)} + KCN_{(excess)} \rightleftharpoons$ complex ion Q

and find the value of $\frac{(x+y)}{2}$ for complex ion P and Q. Where

x = total number of electron(s) present in t_{2g} orbitals of central metal cation of complex ion 'P'

y = total number of electron(s) present in d_{z^2} orbitals of central metal cation of complex ion 'Q'

47. The sum of the molecular weights of nitrogen containing products of given reactions. (Given atomic weight : N = 14, O = 16, H = 1, S = 32, C = 12, Cl = 35.5 and I = 127.00)

I) $NaNO_2 + KI \xrightarrow{H^+}$

II) $NaNO_2 + Zn + KOH \xrightarrow{\Delta}$

III) $NaNO_{2(aq)} + NH^+_{4(aq)} \xrightarrow{\text{boiling}}$

IV) $NaNO_2 + \text{conc. } H_2SO_4 \rightarrow$

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated <u>according to the following marking scheme :</u

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

48. Match the following Column – I with Column – II

	COLUMN-I		COLUMN-II
A)	$Al(OH)_3\ Sol$	**P)**	Negatively charged colloidal particle
B)	$As_2S_3\ sol$	**Q)**	Positively charged colloidal particles
C)	Gold Sol	**R)**	Lyophillic Sol
D)	Starch	**S)**	Lyophobic Sol

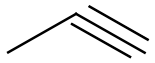
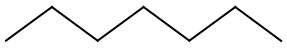
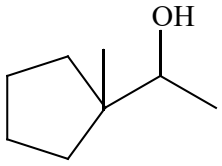
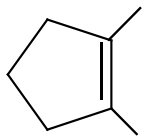
A) A – QS; B – PS; C – PS; D – PR

B) A – PQ; B – QR; C – S; D – RS

C) A – QR; B – S; C – PR; D – Q

D) A – PR; B – RS; C – Q; D – PS

49. Match each process given in Column – I with one or more effect/s in Column – II

	COLUMN-I (Reactant)		COLUMN-II (Product)
P)	 $\xrightarrow[(2)\ \text{conc. } H_2SO_4\ /\ \Delta]{(1)\ \text{dil. } H_2SO_4\ /\ HgSO_4}$	**1)**	1, 3-Dimethyl benzene
Q)	 $\xrightarrow[(2)\ Me-Cl(1\ eq.)\ /\ AlCl_3\ /\ \Delta]{(1)\ Cr_2O_3\ /\ 773K,\ 10-20\ atm}$	**2)**	Toluene
R)	 $\xrightarrow[(2)\ Se\ /\ \Delta]{(1)\ \text{conc. } H_2SO_4\ /\ \Delta}$	**3)**	1, 2-dimethyl benzene
S)	 $\xrightarrow[(3)\ N_2H_4\ /\ \overset{(-)}{O}H,(4)Se\ /\ \Delta]{(1)\ O_3\ /\ H_2O,\ (2)\ \text{dil. } NaOH\ /\ \Delta}$	**4)**	1, 3, 5-trimethyl benzene

A) P – 4; Q – 1; R – 3; S – 2 **B)** P – 2; Q – 1; R – 3; S – 4

C) P – 4; Q – 2; R – 1; S – 3 **D)** P – 3; Q – 1; R – 2; S – 4

50. Match the complexes given in Column – I with the electronic transitions (mainly responsible for their colours listed in Column – II)

	COLUMN-I		COLUMN-II
A)	MnO_4^-	**P)**	$M \rightarrow MCT$
B)	$\left[Mn\left(H_2O\right)_6\right]^{3+}$	**Q)**	$L \rightarrow MCT$
C)	$\left[Co\left(H_2O\right)_6\right]Cl_2$	**R)**	Spin forbidden $d \rightarrow d$
D)	$Fe_3\left[Fe\left(CN\right)_6\right]_2$	**S)**	Spin allowed $d \rightarrow d$

A) A – R; B – Q; C – P; D – S **B)** A – Q; B – R; C – S; D – P

C) A – Q; B – S; C – R; D – P **D)** A – P; B – S; C – R; D – Q

51. Which description in the second list below can be correctly matched to each element or compound in the first list?

	COLUMN-I		COLUMN-II
A)	$\left[S_2O_8\right]^{2-}$	**P)**	Strong reducing agent oxidized to $S_4O_6^{2-}$
B)	S_2F_2	**Q)**	Reaction with water gives SO_2 and S
C)	$\left(S_2O_7\right)^{2-}$	**R)**	Strong oxidizing agent reduced to SO_4^{2-}
D)	$\left[S_2O_3\right]^{2-}$	**S)**	Contains $S-O-S$ linkage

A) A – R; B – P; C – Q; D – S **B)** A – S; B – P; C – Q; D – R

C) A – S; B – Q; C – Q; D – R **D)** A – R; B – Q; C – S; D – P